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Xiyuan Bao

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Education

Ph.D. in Geophysics, University of California, Los Angeles	2018-Mar 2024
M.S. in Geophysics, University of California, Los Angeles	2018-2020
B.S. in Geophysics with Honors , University of Science and Technology of China	2014-2018
Minor in Computer Science, University of Science and Technology of China	2014-2018

Experience

Gordon and Betty Moore Foundation Postdoctoral Fellow, Harvard University	Oct 2026-
Reginald A. Daly Postdoctoral Fellow, Harvard University	Oct 2024-Sep 2026
Postdoctoral Scholar, University of California, Los Angeles	Mar 2024-Sep 2024
Graduate Research Assistant, University of California, Los Angeles	Sep 2018-Mar 2024
Undergrad Research Assistant, University of Science and Technology of China	Sep 2017-May 2018
Visiting Undergraduate, University of Illinois, Urbana-Champaign	Jun-Aug 2017

Research Interests

- Dynamics of the Solid Earth Interior
- Surface Processes and Coupling with Planetary Interiors
- Experimental, Computational and Machine Learning Methods in Geophysics

Grants, Honors & Awards

- Harvard Data Science Initiative-Amazon Web Service (HDSI-AWS) Impact Computing Project (Lead Proposal Writer, PI ineligible due to postdoctoral status, $209,302) — 2026-2028
- Gordon and Betty Moore Foundation Postdoctoral Fellowship ($192,308) — 2026-2028
- Reginald A. Daly Postdoctoral Research Fellowship, Harvard University ($171,500) — 2024-2026
- Study of the Earth's Deep Interior Section Award for Graduate Research, AGU — 2023
- Computers and Geosciences Research Scholarship, IAMG — 2022
- Jiuzhang Zhao Talent Program in Earth and Space Sciences, USTC — 2015-2018
- Award from National Undergraduate Innovation Training Program, USTC — 2017-2018
- Earth Science Climbing Scholarship, USTC — 2015, 2017
- Silver Scholarship for Outstanding Students, USTC — 2016
- First Prize of Physics Innovation Research Experiment Thesis Competition, USTC — 2016

Publications

* denotes publications for which I served as the (co-)corresponding author;
† denotes students or postdocs (co-)advised by me.

[9] **Bao, X.***, Wamba, M. D. & Stracke, A. A joint geophysical-geochemical deep-mantle zoning map beneath East Africa and the Indian Ocean. ***Science*** **393,** 684–689. https://www.science.org/doi/full/10.1126/science.aea5466 (2026).

[8] Coulson, S., Lloyd, A., **Bao, X.,** Mitrovica, J. X., Dangendorf, S., Pan, L., Valencic, N., Tamisiea, M. E., Al-Attar, D. & Heathcotte, D. Inverting Sea Surface Height Data Yields Greenland Ice Mass Changes (1993-2019): A Proof of Concept. *Geophysical Journal International.* https://academic.oup.com/gji/advance-article/doi/10.1093/gji/ggag187/8677305 (2026).

[7] Lucas, E., Richards, F., Cederberg, G., **Bao, X.,** Hoggard, M. J., Tsuji, S. R. J., Latychev, K., Tsuji, L. J. S. & Mitrovica, J. Emergence of Antarctic mineral resources in a warming world. ***Nature Climate Change***. https://www.nature.com/articles/s41558-026-02569-1 (2026).

[6] **Bao, X.*** & Lithgow-Bertelloni, C. R. Self-correction of the optical distortion effect of thermal plumes in particle image velocimetry. *Physics of Fluids.* https://doi.org/10.1063/5.0233759 (2024).

[5] **Bao, X.*.** Giant impact and Earth's mysterious blobs: An interdisciplinary revelation. *Science Bulletin* **69,** 293–294. https://cdn.sciengine.com/doi/10.1016/j.scib.2023.12.028 (2024).

[4] **Bao, X.*,** Mittal, T. & Lithgow-Bertelloni, C. R. Determining Mid-Ocean Ridge Geography from Upper Mantle Temperature. *Earth and Planetary Science Letters* **641,** 118823. https://www.sciencedirect.com/science/article/pii/S0012821X24002565 (2024).

[3] Zhang, X., Brown, E. L., Zhang, J., Lin, J., **Bao, X.** & Sager, W. W. Magmatism of Shatsky Rise controlled by plume-ridge interactions. ***Nature Geoscience***. https://www.nature.com/articles/s41561-023-01286-0 (2023).

[2] Li, S., Li, J., Ferrand, T. P., Zhou, T., Lv, M., Xi, Z., Maguire, R., Han, G., Li, J., **Bao, X.,** *et al.* Deep geophysical anomalies beneath the Changbaishan Volcano. *Journal of Geophysical Research: Solid Earth,* e2022JB025671. https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2022JB025671 (2023).

[1] **Bao, X.*,** Lithgow-Bertelloni, C. R., Jackson, M. G. & Romanowicz, B. On the relative temperatures of Earth's volcanic hotspots and mid-ocean ridges. ***Science*** **375,** 57–61. ISSN: 0036-8075. https://www.science.org/doi/10.1126/science.abj8944 (2022).

MANUSCRIPTS SUBMITTED

[7] Wurmser, S. S., Al Asad, M., **Bao, X.,** Creveling, J., Mitrovica, J. X. & Tryon, C. A. *Reopening the Southern Gate: Early Pleistocene Sea Level and Hominin Dispersal Across the Red Sea*

[6] Acquadro, J.†, **Bao, X.*** & Deng, J. *Machine-Learned Local Environments Reveal Framework Controls on MgSiO3 Liquid-Crystal Similarity* Under review at *PNAS NEXUS*.

[5] Powell, R., Cederberg, G., Dangendorf, S., Lucas, E., **Bao, X.,** Creveling, J., Tsuji, S., Tsuji, L. & Mitrovica, J. X. *Disproportionate sea level rise in small island developing states of the Pacific and Indian Oceans driven by Greenland and Antarctic ice mass loss* Under review at ***Communications Earth & Environment***.

[4] Cai, R., **Bao, X.,** Foley, S. F., Giuliani, A., Jiang, S., Pearson, D. G. & Liu, J. *Evidence for a highly fusible fundamental constituent in the convective mantle.* In Revision.

[3] **Bao, X.*** & Lithgow-Bertelloni, C. R. *A method for measuring and analyzing four-dimensional flow fields viscous in Rayleigh-Bénard Convection* Revision Under Review.

[2] Wang, W., Zhang, X., Xu, F., Cheng, Z., **Bao, X.,** Stracke, A., Zha, C., Zhan, F. & Lin, J. *Ultra-depleted mantle beneath the Marion Rise and its implications for ocean rise uplift*

[1] **Bao, X.*** & Lithgow-Bertelloni, C. R. *Morphology, interactions, and evolution of laminar thermal plumes revealed by four-dimensional velocity measurements and analysis: Application to Earth's mantle* Revision Under Review.

MANUSCRIPTS IN PREP

[2] **Bao, X.*,** Coulson, S., Valencic, N., Mousavi, M., Dangendorf, S., Lloyd, A. & Mitrovica, J. *Neural Operators Reveal Scale-Dependent Limits on Recovering Greenland Ice Loss from Sea-Level Fingerprints*

[1] **Bao, X.*,** Cao, Z., Gourley, K. C., Gutierrez, S., Liang, Y., Zhou, D., Buffett, B. A., Mittal, T., Li, J., Pierrehumbert, R., Schaefer, L., Orman, J. V. & Foley, B. J. *Coupled Evolution of Earth's Hydrogen Distribution and Thermal History.*

ORAL PRESENTATIONS

[19] ***Invited:*** *Deep Provenance of East African and Indian Ocean Hotspots through Tomography, Graph Theory, and Geochemical Cross-validation* Geodynamics Seminar (Lamont-Doherty Earth Observatory, Palisades, NY, US, Feb. 2026).

[18] ***Invited:*** *What Powers Earth's Hotspot Volcanism? What Does It Reveal About the Deep Mantle?* Geo. Sci. Special Seminar (University of Chicago, Chicago, IL, US, Feb. 2026).

[17] *Inverting Sea Level Fingerprints for Ice Mass Changes Using Neural Operators: How Much Can We Recover and How Unique Is It?* AGU Fall Meeting 2025 (New Orleans, LA, US, Dec. 2025).

[16] ***Invited:*** *Deep Provenance of East African and Indian Ocean Hotspots through Tomography, Graph Theory, and Geochemical Cross-validation* Brownbag Seminar (Princeton University, Princeton, NJ, US, Nov. 2025).

[15] ***Invited:*** *Deep Provenance of East African and Indian Ocean Hotspots through Tomography, Graph Theory, and Geochemical Cross-validation* Geophysics Hour (Arizona State University, Tempe, AZ, US, Nov. 2025).

[14] ***Invited:*** *Plumes over LLSVPs: Supercharging multi-scale flow resolution with hybrid experimental-adjoint digital twins* Geophysics Seminar (Brown University, RI, US).

[13] *Plumes over LLSVPs: Supercharging multi-scale flow resolution with hybrid experimental-adjoint digital twins* BiSEPPS Seminar (Harvard University, MA, US, Nov. 2024).

[12] ***Invited:*** *A Song of Ice and Fire: Operator Learning for Viscous Flow from Mantle Convection to Ice Dynamics* Daly Special Seminar (Harvard University, MA, US, Feb. 2024).

[11] ***Invited:*** *Hotspots from Top to Bottom* Geophysics Seminar (University of Science and Technology of China, Anhui, China, Jan. 2024).

[10] ***Invited:*** *Illuminating Mantle Convection: Unraveling Plume Dynamics over LLSVPs in the Laboratory* AGU Fall Meeting 2023 (San Francisco, CA, US, Dec. 2023).

[9] ***Invited:*** *Hotspots from Top to Bottom* Earth Sciences Seminar (University of Southern California, CA, US, Oct. 2023).

[8] *The plume zoo of thermal LLSVPs* AGU Fall Meeting 2022 (Chicago, IL, US, Dec. 2022), V26A–02.

[7] *Coupled Evolution of Earth's Hydrogen Distribution and Thermal History* AGU Fall Meeting 2022 (Chicago, IL, US, Dec. 2022), MR33A–06.

[6] ***Invited:*** *Hotspots from Top to Bottom* Seismo Lab Seminar (California Institute of Technology, CA, US, Nov. 2022).

[5] ***Invited:*** *Are hotspots hotter than ridges?* IGCP 648 Virtual Seminar Series (Curtin University, Australia, Nov. 2022), Series 5, Seminar 5. https://www.youtube.com/watch?v=30B21c7NGIY.

[4] *Experimental Investigation of Purely Thermal LLSVPs* UCLA EPSS Seismology and Tectonics Seminar (UCLA, CA, US, May 2022).

[3] *Accurate Prediction of Ocean Basins Using Upper Mantle Potential Temperatures* AGU Fall Meeting 2021 (New Orleans, LA, US, Dec. 2021), DI34A–05.

[2] *Accurate Prediction of Ocean Basins Using Upper Mantle Potential Temperatures* UCLA EPSS Geology/Geophysics Seminar (UCLA, CA, US, Oct. 2021).

[1] *Correlation of geochemical signals with MORB and OIB temperatures* UCLA EPSS Geocheminar (UCLA, CA, US, Oct. 2019).

Poster Presentations

[13] **Bao, X.,** Wamba, M. D. & Stracke, A. in. AGU Fall Meeting 2025 (New Orleans, LA, US, Dec. 2025).

[12] **Bao, X.,** Coulson, S., Valencic, N., Mitrovica, J., Dangendorf, S., Mousavi, M. & Lloyd, A. in. 2025 Gordon Research Seminar: Interior of the Earth (June 2025).

[11] **Bao, X.** & Lithgow-Bertelloni, C. R. in. AGU Fall Meeting 2024 (Washington DC, US, Dec. 2024).

[10] **Bao, X.** & Lithgow-Bertelloni, C. R. in. 2024 Ada Lovelace Workshop on Modelling Mantle and Lithosphere Dynamics (University of California, Berkeley, CA, US, Sept. 2024).

[9] **Bao, X.** in. AGU Fall Meeting 2023 (San Francisco, CA, US, Dec. 2023).

[8] **Bao, X.** & Lithgow-Bertelloni, C. R. in. 2023 Gordon Research Seminar: Interior of the Earth (Mount Holyoke College, MA, US, June 2023).

[7] **Bao, X.** & Lithgow-Bertelloni, C. R. in. Seismo Lab Centennial (California Institute of Technology, US, Nov. 2022).

[6] **Bao, X.** & Lithgow-Bertelloni, C. R. in. 2022 Ada Lovelace Workshop on Numerical Modelling of Mantle and Lithosphere Dynamics (Hévíz, Hungary, Aug. 2022).

[5] **Bao, X.** & Lithgow-Bertelloni, C. R. in. CIDER2022 (University of California, Berkeley, CA, US, July 2022).

[4] **Bao, X.,** Lithgow-Bertelloni, C. R. & Jackson, M. G. in. AGU Fall Meeting 2020 (San Francisco, CA, US, Dec. 2020), DI007–0006.

[3] Wang, Y., Liu, L. & **Bao, X.** in. AGU Fall Meeting 2020 (San Francisco, CA, US, Dec. 2020), DI005–0005.

[2] **Bao, X.,** Lithgow-Bertelloni, C. R. & Jackson, M. G. in. AGU Fall Meeting 2019 (San Francisco, CA, US, Dec. 2019), DI41D–0028.

[1] **Bao, X.** & Lithgow-Bertelloni, C. R. in. 2019 Gordon Research Seminar: Interior of the Earth (Mount Holyoke College, MA, US, June 2019).

Professional Service

- **Guest Editor**, *PNAS Nexus*, First Call for Papers, in *Machine Learning and Geosciences*
- **Primary Session Convener and Chair**, *Advances in Machine Learning for Solid Earth Geoscience*, AGU Fall Meeting 2025
- **Primary Session Convener and Chair**, *Advances in Deep Earth–Surface Interactions*, AGU Fall Meeting 2024
- **Discussion Leader**, *The Structure and Composition of the Lower Mantle*, Gordon Research Seminar 2023
- **Primary Session Convener and Chair**, *Advances in Mantle Convection and Planetary Evolution*, AGU Fall Meeting 2022
- **Reviewer**, *Nature Geoscience*; *Science Advances*; *Journal of Geophysical Research: Solid Earth*; *Geophysical Journal International*; *Planetary Science Journal*; *Physics of Fluids*; *Earth and Planetary Physics*; *Frontiers in Earth Science*
- **Reviewer**, National Science Foundation, Geophysics Program

Other Service

- Committee Member of UCLA EPSS Family Mentorship Program (EFMP)

Outreach

- *Rheology, Plate Tectonics with Food* Saturday Science Academy, UCLA/Charles R. Drew University Mar 2023
- *Why Hawaii is a volcano? How do Alaskan volcanos differ?* EYU, UCLA Nov 2022
- **Media Interview** about our *cold hotspots, e.g. ScienceNews, New Scientist*[1,2] *Popular Science, and Deutschlandfunk* Jan 2022
- **Invited Outreach Article** *Rocks beneath volcanic hotspots can be surprisingly cool* Jan 2022
- *The sweet smell of Earth's mantle: Why is Hawaii a volcano?* EYU, UCLA Nov 2020, 2021
- Lab visit from Tokyo Tech. UCLA Feb 2020
- *Why is Hawaii a volcano?* Explore Your Universe, UCLA Nov 2019

Teaching Experience

- Blue Planet: Introduction to Oceanography, Teaching Assistant, UCLA Spring 2020
- Earthquakes, Teaching Assistant, UCLA Spring 2019, Fall 2021
- AI for Earth and Planetary Sciences, Guest Lecturer, Harvard University Spring 2026

Mentoring Experience

- Natasha Valencic, PhD student at Harvard, *Separating Decadal Ice-Melt Fingerprints from Stereodynamic Sea-Level Change with Machine Learning* June 2026 -
- Doyoon Park, PhD student at Princeton, *Viscosity of Davemaoite Under Lower Mantle Conditions from Surrogate Machine Learning Molecular Dynamics* Nov 2025 -
- Jake Acquadro, PhD student at Princeton, *Machine-Learned Local Environments Reveal Framework Controls on MgSiO3 Liquid-Crystal Similarity* Nov 2025 -
- Kavya Agarwal, visiting undergrad at UCLA, *Compositional and thermodynamical effect on mantle plumes;*
  Critical plume merge distance with shadowgraph; now PhD student at Yale Sep 2021 - Aug 2022
- Wan Ki (Arthur) Lo, undergrad at UCLA, *Dynamics of Adjacent Mantle Plumes through Physical and Computer Simulations*, won *2021 Harold and Mayla Sullwold Scholarship*; now PhD student at PSU Jan 2020 - Jun 2022
- Jade Wight, undergrad at UCLA, *Modeling Mantle Plumes and Their Effect on Dynamic Topography*, won *2020 Deane Oberste-Lehn Scholarship*; past PhD student at UHawaii and now geologist at AECOM Oct 2019 - Jun 2021